Ken (Tongli) Su

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CS PhD student working on **LLM agents**, with **alignment and safety** as the long-term direction. Current work gives coding agents a **world model** of what a change will break before it runs, **memory** that stays reachable over long horizons, and **repository-graph retrieval** inside their own search loop. 2+ years of prior industry ML engineering (GenAI, retrieval, data pipelines).

Education

Emory University — Atlanta, GA
Ph.D. in Computer Science (Advisor: Ali Emami) — Aug 2025 – Dec 2028 (expected)

Columbia University — New York, NY
M.S. in Data Science, GPA: 3.95/4.00 — Dec 2023

Emory University — Atlanta, GA
B.A. in Computer Science & B.A. in Economics (Dual Major), GPA: 3.84/4.00 — May 2022

Selected Publications & Research

[1] **T. Su**[*], Y. Hu[*], L. Zhao, B. Zhu, J. Somasundaram, H. Haque. **"Software World Models: From Consequence Prediction to Decision Value."** *arXiv:2610.04940*, 2026. *(Under review; [*]equal contribution.)*

- Formulated **system-level consequence prediction** (forecast the **blast set** a change will break before it runs) and trained the world model with **GRPO** on executed outcomes: **32% higher** blast-set F1 and **59% lower** ranking regret than static reachability; **35% higher** agent return on held-out real libraries, outperforming Meta CWM and Qwen-AgentWorld.

[2] Y. Hu, **T. Su**, Y. Sun, S. Ma, Q. Tan, L. Zhao. **"Adaptive Memory Index Evolution for Agents."** *arXiv:2610.34977*, 2026. *(Under review.)*

- **AMIE**: a plug-in layer that distills an LLM teacher's keep/suppress judgments into a **Qwen3-8B LoRA** policy that evolves per-memory access state; improved F1/accuracy in **all 15** comparisons (**5 memory systems** × **3 benchmarks**) by up to **25%**.

[3] Y. Hu[*], **T. Su**[*], L. Zhao, B. Zhu, H. Haque. **"LARGER: Lexically Anchored Repository Graph Exploration and Retrieval."** *arXiv:2605.16352*, 2026. *(Under review; [*]equal contribution.)*

- Anchored a coding agent's lexical search hits in a repository graph and expanded confidence-filtered neighborhoods (imports, call chains, code–test links) *inside* its search loop with no new actions; built the **AST-based graph construction**.
- **+13.9 Acc@5** and **+12.2 Recall@5** over the strongest of **7 baselines** (incl. SWE-agent, OpenHands, Claude Code, Codex) across **4 benchmarks** spanning localization, test generation, and codebase QA.

[4] **T. Su**, A. Rafiei, M. van Assen, R. Sameni, G. D. Clifford, F. Marzbanrad, N. Katebi. **"Cross-Modal Generative Framework for Signal Translation from Fetal–Maternal Electrocardiograms to Fetal Doppler Waveforms."** *IEEE EMBC 2026* (**Oral**).

[5] **T. Su**, N. Barari, N. Katebi. **"A Review of Multimodal Pretraining Strategies for Physiological Wearable Sensor Foundation Models."** *Under review*, 2026.

Industry Experience

Research Intern | Causal Dynamics Lab — Seattle, WA | May 2026 – Aug 2026

- Built a **software world model** for multi-repository microservice systems that predicts which downstream services, libraries, and datastores a change will break *before* it reaches CI/CD, catching failures that local tests miss.
- Engineered a sandboxed **execution harness** that restores frozen baselines, applies a patch, and reruns unit, service-health, and end-to-end checks across Go/Node/Python **gRPC microservices** and real libraries (Marshmallow, OpenTelemetry).
- Built the training and serving pipeline (dependency graphs as LLM graph tokens, **SFT + GRPO** on executed outcomes) and a **pre-merge risk check** that runs costly integration checks only when needed: ∼**40% fewer** services broken on held-out systems.

Machine Learning Engineer, GenAI | DuPont Specialty Products USA, LLC — New York, NY | Jul 2024 – Aug 2025

- Built a lead identification pipeline on **Azure Databricks** (Spark, Delta Lake, generative models) behind an internal sales co-pilot that became the company's primary prospecting tool (**$60M+** estimated value); fine-tuned **Sentence-BERT** with triplet loss for retrieval and ranking: **Recall@5 0.86**, **MRR 0.64**, **+27%** over the incumbent GPT-based system with hallucinations eliminated.
- Engineered an async, load-balanced **OpenAI API** enrichment module: runtime 10 h → 6 h (**40%**), saving **$12K/yr** in token costs.
- Turned user feedback into model and feature improvements in a six-week sprint: **DAU +48%**, satisfaction 4.0 → 4.6/5.

Technical Skills

Agents & LLMs	LLM & coding agents, world models, agent memory, agent evaluation, RL post-training (SFT, GRPO), LoRA, distillation, retrieval & ranking (RAG, BM25, graph retrieval), Hugging Face, Sentence-BERT, OpenAI API
ML & Deep Learning	PyTorch, Scikit-learn, XGBoost; graph neural networks (R-GCN), attention, cross-modal & generative models, self-supervised pretraining, time-series & biosignals
Languages & Infra	Python, SQL, C++, Bash; Spark, PySpark, Pandas, Databricks, Delta Lake, PostgreSQL; AWS, Azure, GCP

Service & Honors

Reviewer, **NeurIPS 2026** · Oral presentation, **IEEE EMBC 2026** (first author)